

① Design Uber / Lyft.

I. Functional Requirement

1. User should be able to find out cabs around them.
2. User should be able to book the cab.
3. The app should be able to track location of the driver.

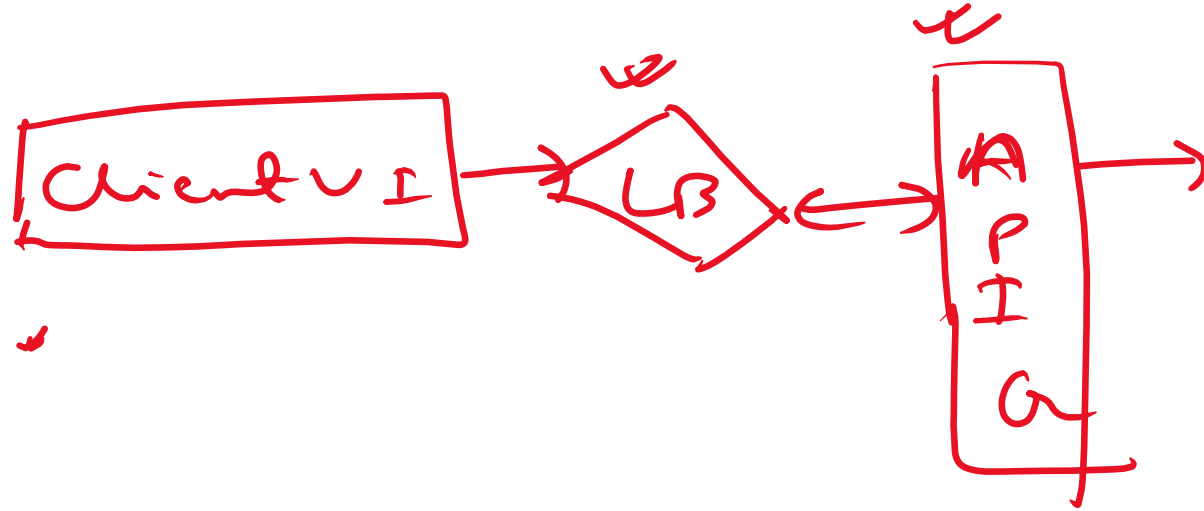
II. Non functional Requirement

1. Scalability.
2. Consistency.
3. Highly Available
4. Fault Tolerant
5. Low Latency.

III. Capacity Estimation :-

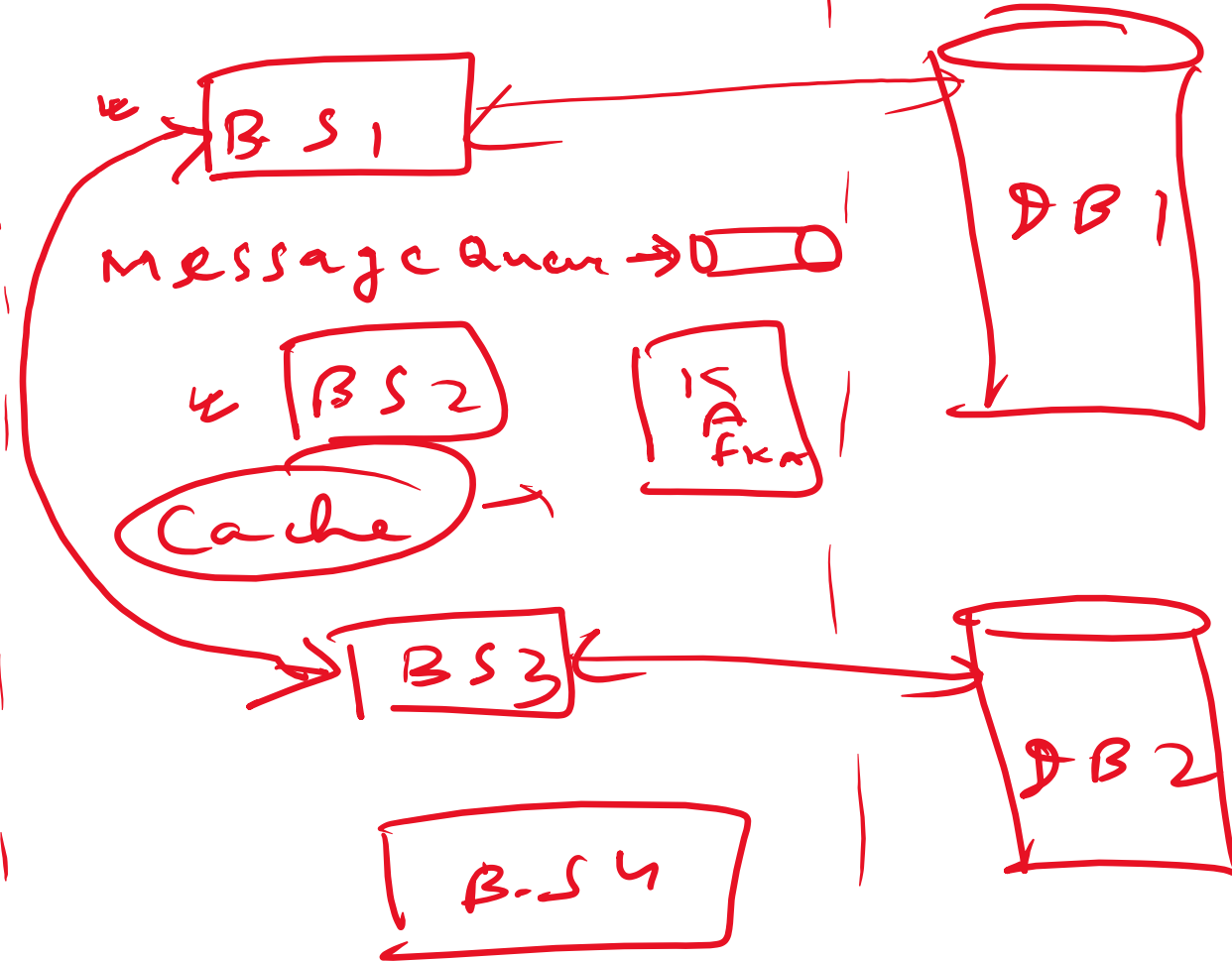
1. DAU :- ~ 15 millions users
2. MAU :- ~ 100 million users.

○



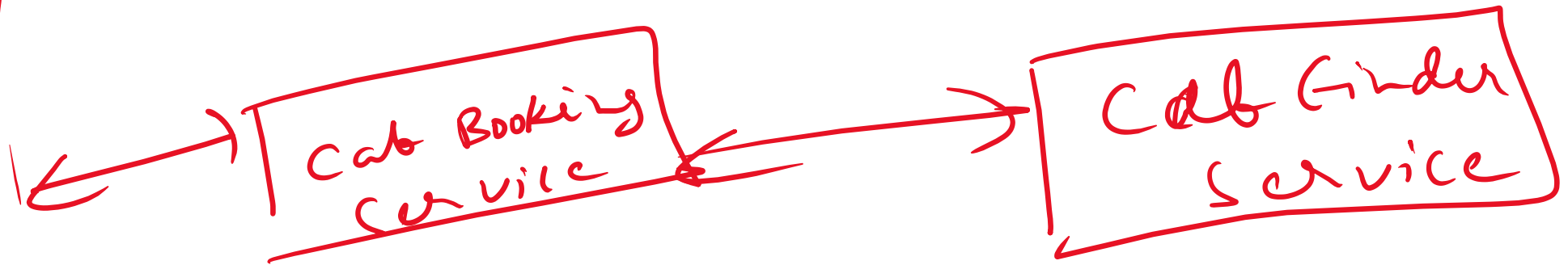
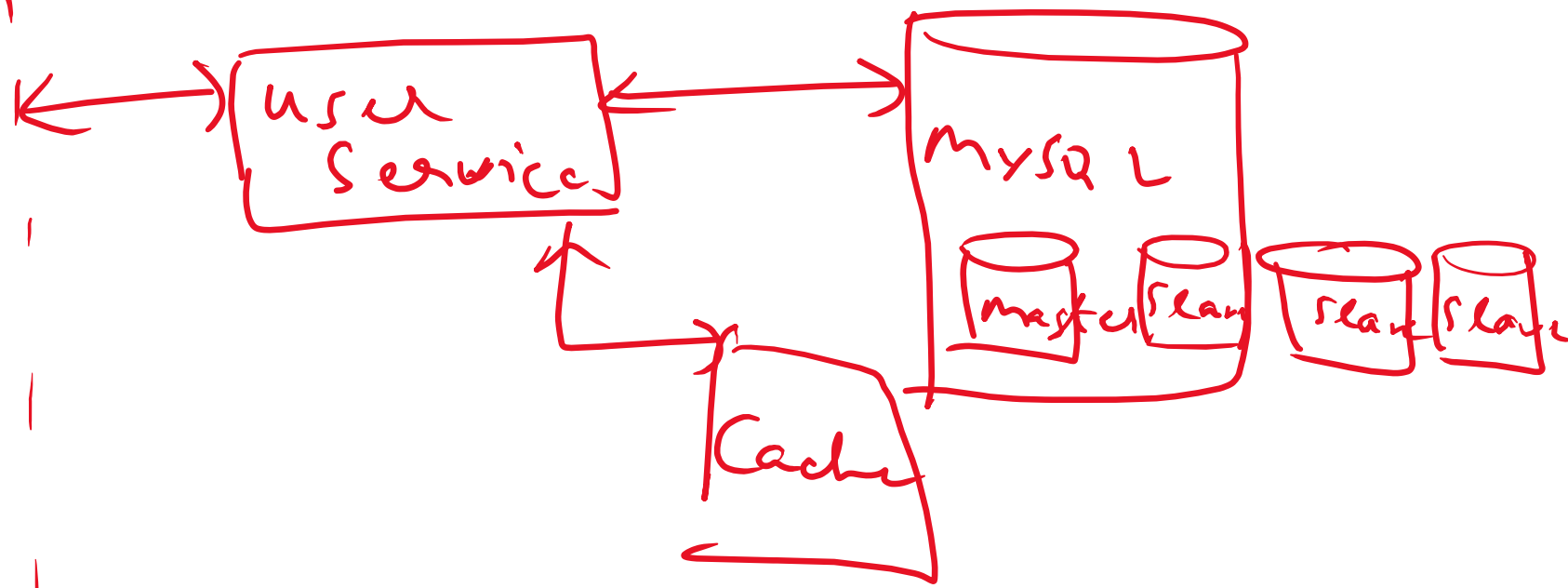
BACKEND SERVICES

DATA BASE



① Backend Services :-

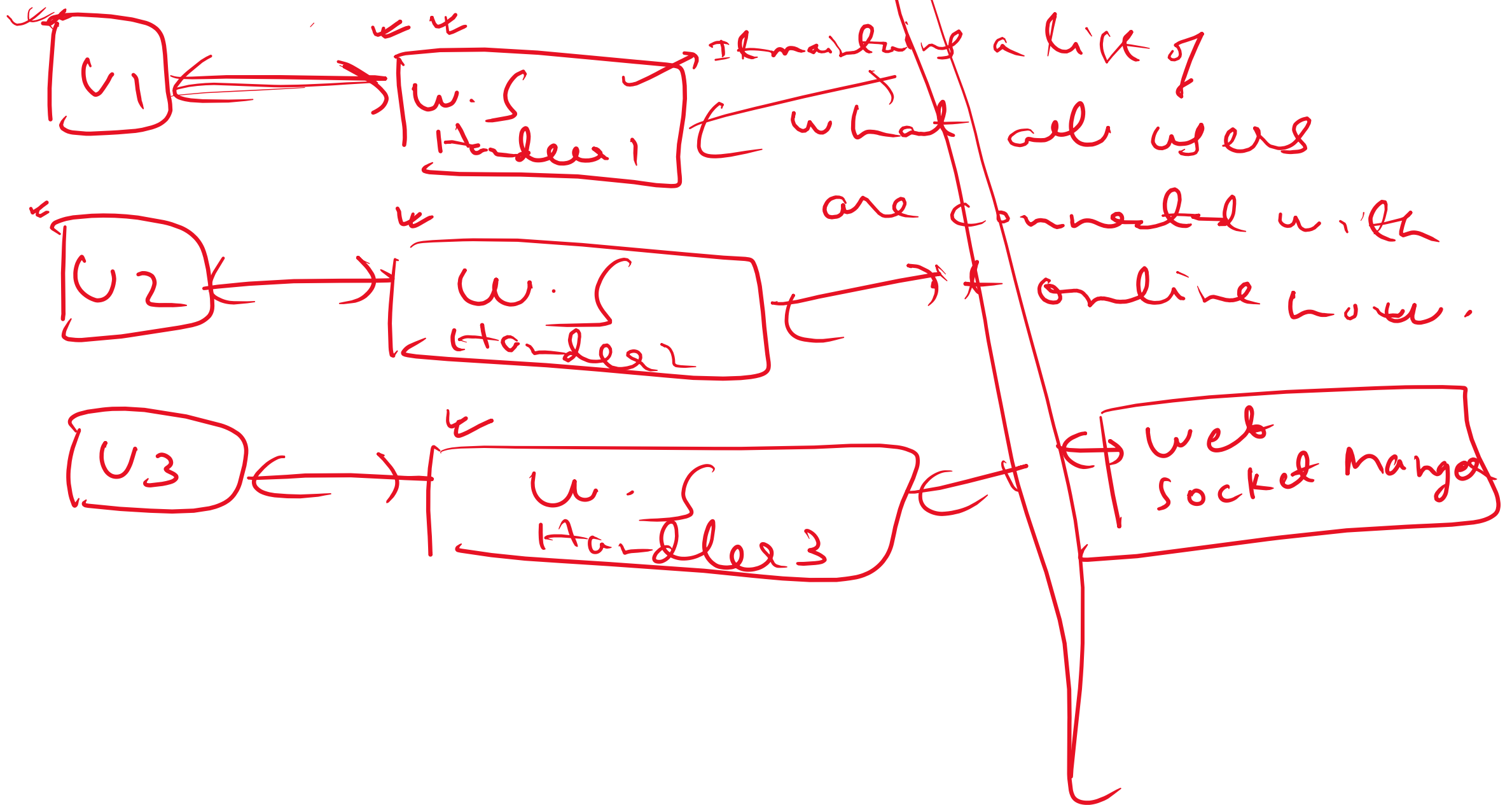
1. User Service :- All user or customer who wants to book a cab will get connected via LB and API gateway and communicate to User Service.
So User Service is a repository of all user information.



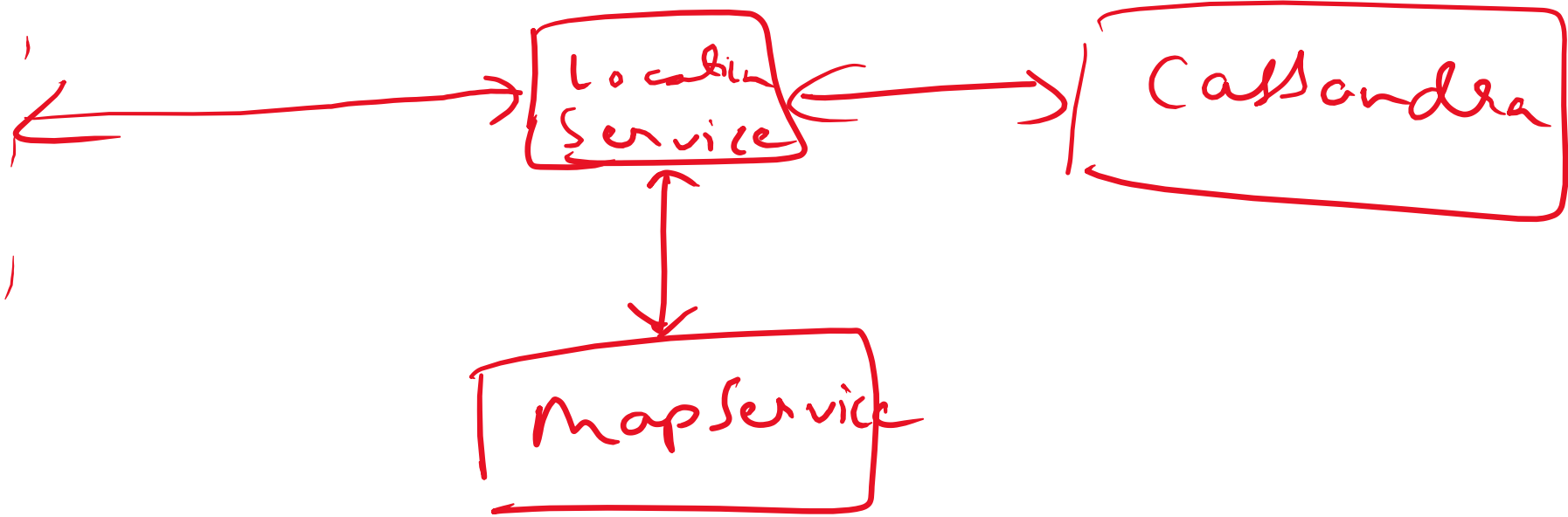
2. Cab Booking Service^(CBS):- When customer wants to book the cab, in that situation, it is CBS which will receive the request from client. It makes a web socket connection with the user's app which would display them few cabs on their UI which are around them. So it would communicate with other Service like Cab Finder Service.

and send back the user stating
the cab is booked and these are details
of cab driver.

① Driver Service:- It will be
repository of information related
to driver.

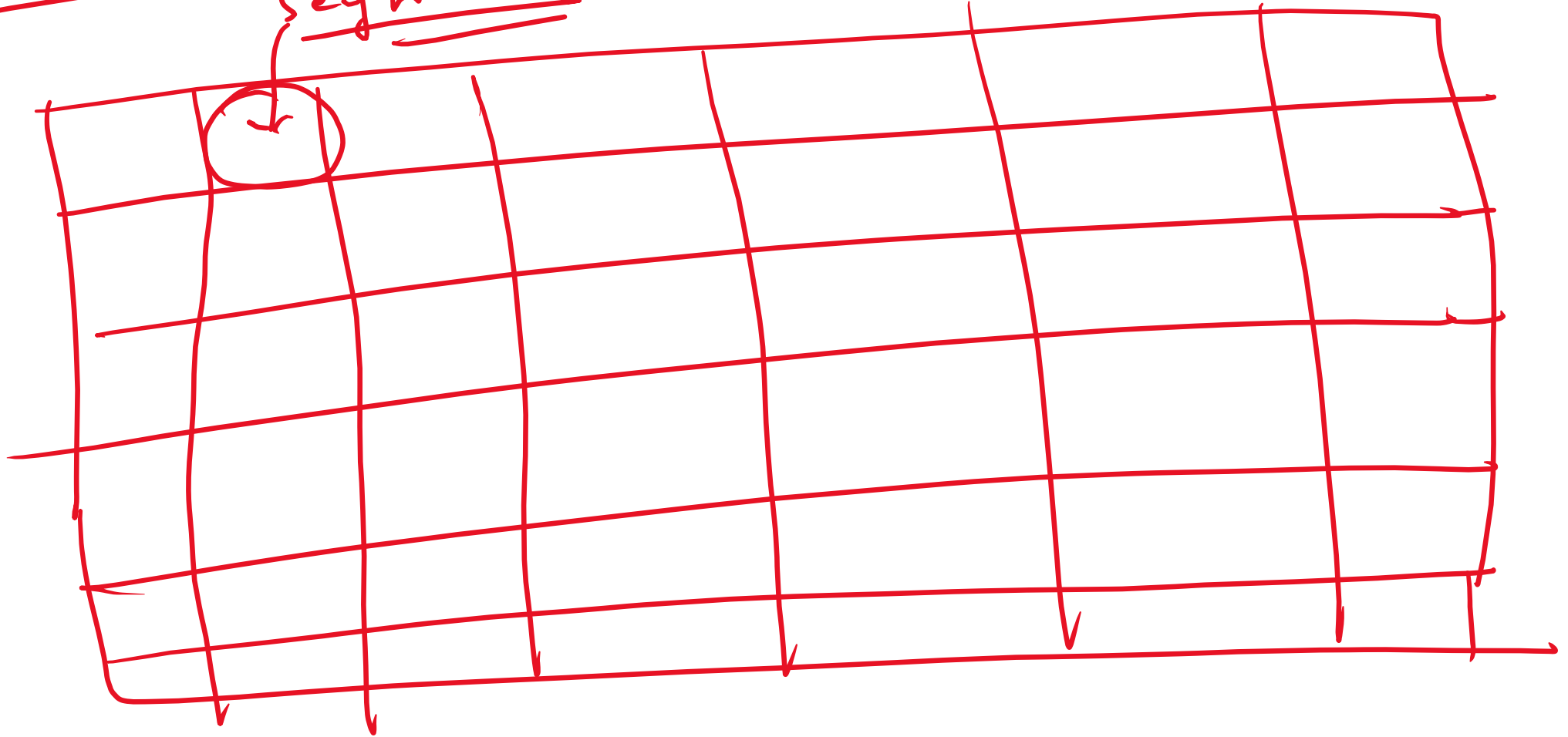


① Location Service :- As the driver will send ping to the server about it's new location say every 20 second whenever it is moving, so that thing will be managed by location Service. So it will store the information related to location of a driver. It will communicate to map service to find out which segment the driver and user belongs to.

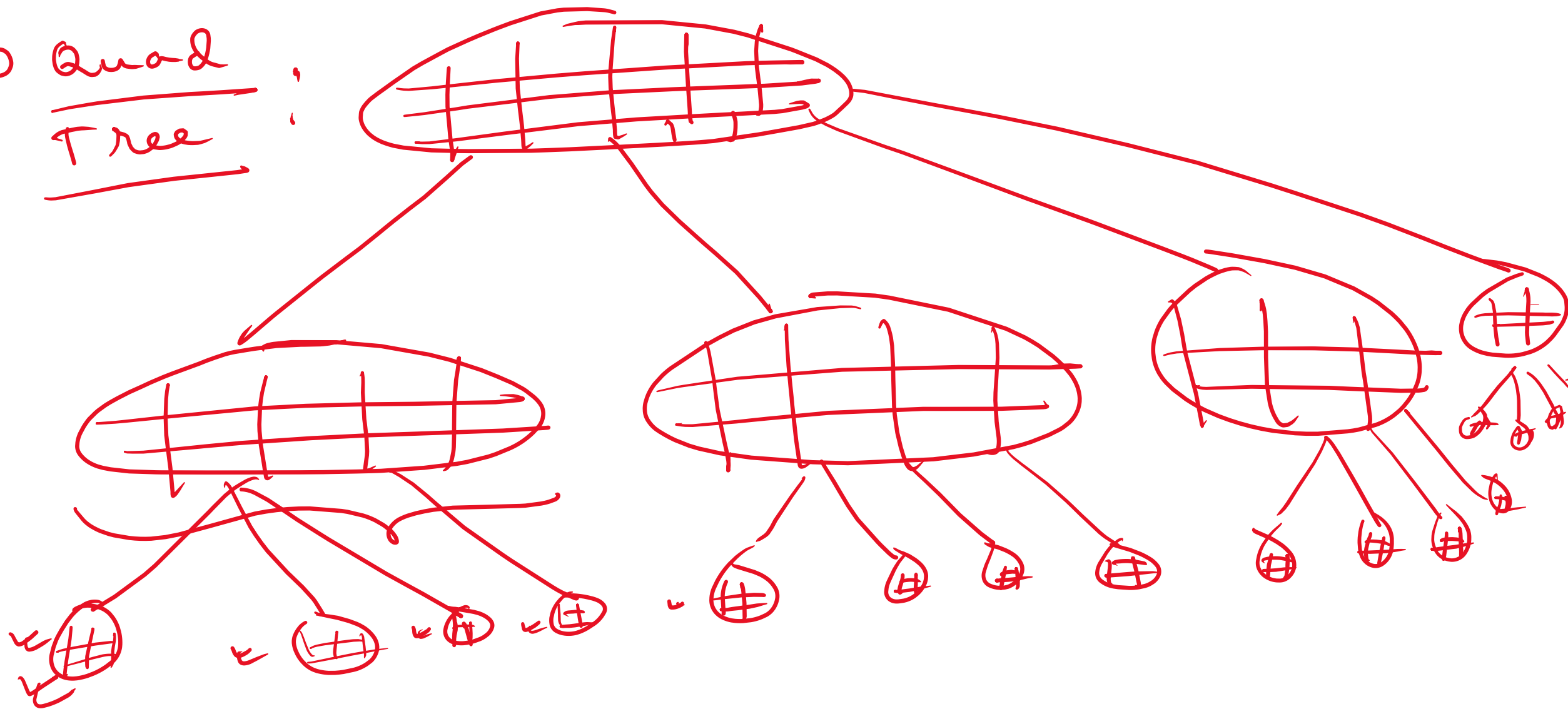


① Map Service :-

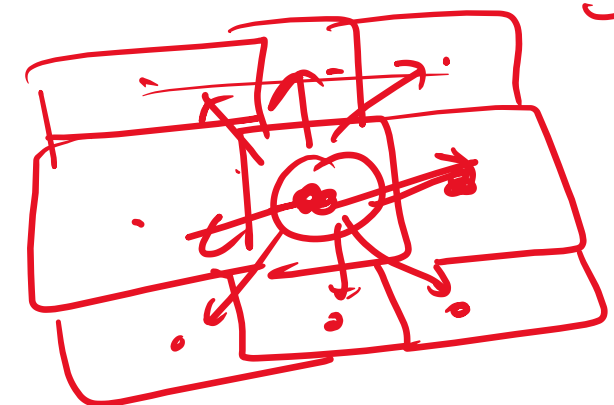
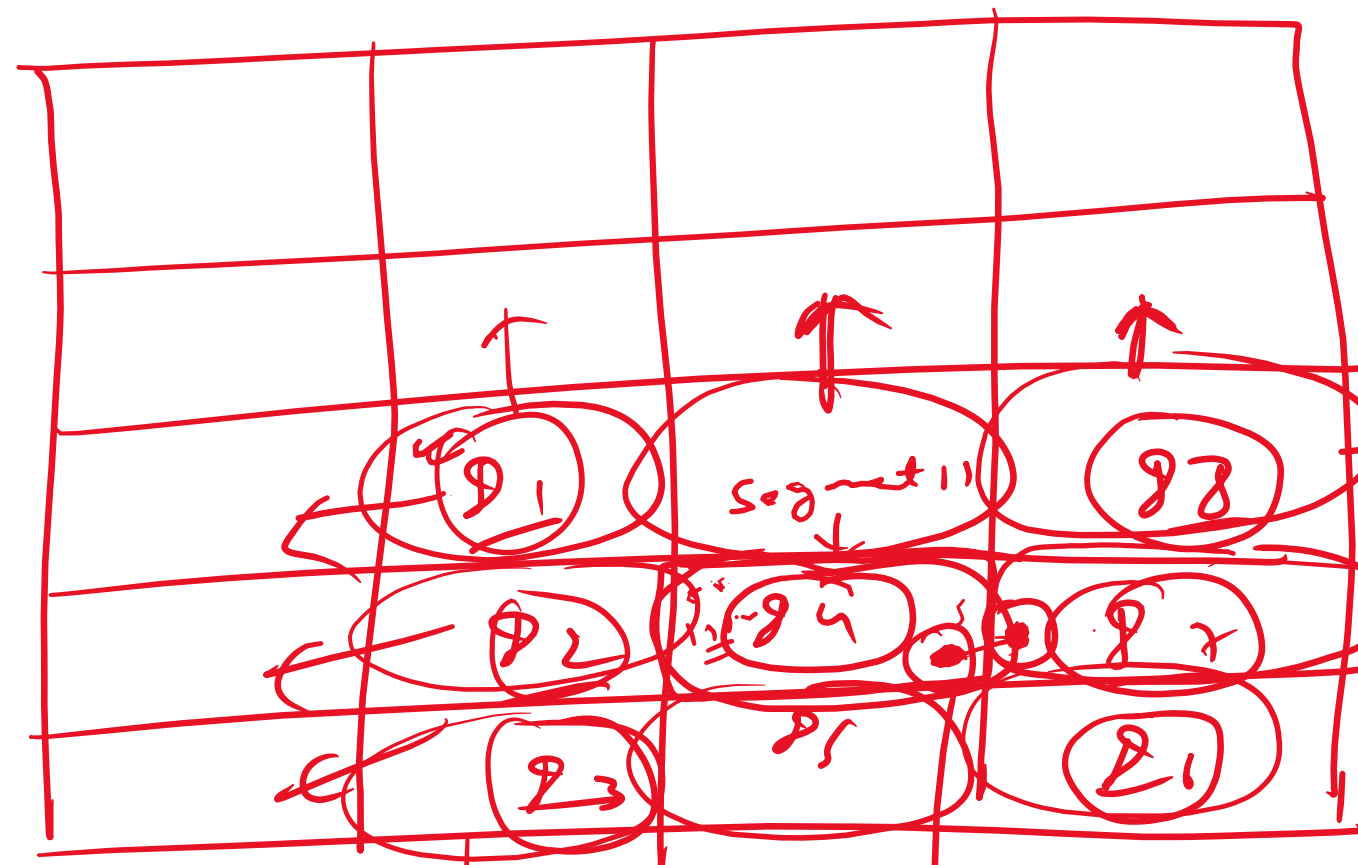
Segment



① Quad
Tree



BFS



Customer